

Smart contract security review

sasa v6 · multi-chain launchpad contracts · Robinhood Chain and Base

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Review	Three rounds of independent external review, plus sasa's own test suites
Code reviewed	commit bdb63168671fe6ad7da059b2868568d683d12939, with the round 2 and round 3 fixes applied on top
Result	No critical, high or open medium findings. All findings were fixed or accepted with reasons; the reviewer judged the code ready for production after round 3.

This document is written and published by sasa. It summarizes an independent review in our own words; it is not a certificate issued by the reviewer. A review lowers risk but cannot prove the absence of bugs, and it does not cover third-party systems.

1. Scope

About 2,100 lines of Solidity (0.8.26, via IR, EVM cancun), the contracts that hold or move users' money:

Contract	Role
OmniFactory	Launches a coin and its curve on each chain at the same address (CREATE3); beta money cap; pause of buys
OmniHub	Cross-chain race: freeze, report, decide, settle or reopen (LayerZero); single-chain coins decided in place
UsdCurveV6	Bonding curve per chain priced in dollars, 1% fee (30% creator)
MigratorV6 / MigratorV6Core	Opens the Uniswap v4 pool at graduation with liquidity locked forever; 2% graduation fee; buyback
LaunchCoin	The coin: 1B cap, moves only after graduation and only to the winning chain, optional creator lock
ConsolidatorV6, AcrossUsdBridge	Moves a losing chain's money to the winner through Across
OmniOrders	Take profit, stop loss and buy-the-dip orders, never filled outside the owner's limits
UsdPoolRouter, NeuronGraduationHook	Trading in graduated pools; only sasa can create a coin's pool
UsdFeeSplitter, SasaBoost, deployers	Fee split, paid placement, CREATE3 deployers

Out of scope: earlier contract versions kept in the repository, testnet-only contracts, deployment scripts, the website and off-chain services.

2. How the review was done

- Round 1: full manual review of the scoped contracts against a written brief (architecture, roles, invariants, trust assumptions).
- Round 2: review of the fixes for the round 1 findings, by diff, and a check for regressions.
- Round 3: review of the final change (coin moves after graduation) and a final verdict.

- Alongside: sasa's unit tests (71 for these contracts) and tests on copies of Robinhood Chain and Base mainnet with the real Uniswap v4, USDC, USDG and Across contracts.

3. Findings and resolutions

ID	Severity	Finding	Resolution
M-1	Medium	After graduation anyone could move a holder's coins to the winning chain at a moment the holder did not choose.	Fixed. For 24 hours only sasa's keeper may move other holders; each holder can move their own coins any time; after 24 hours anyone may, so coins never get stuck. Moves always go to the holder's own address.
M-2	Medium	If the bridge ever refunded less than a deposit, that money could not be re-sent.	Fixed. The owner can re-send the refunded amount, once, only to the same coin and chain.
L-1	Low	Rounding dust could stay in a settled curve forever.	Fixed. Anything left above what the curve owes goes to the protocol fee at settlement.
L-2	Low	If more money than expected arrives, the pool opens slightly above the graduation price.	Accepted, documented. It only benefits holders.
L-3	Low	An order's upper limit is checked after the trade.	Accepted. A failing check undoes the whole trade.
L-4	Low	If the keeper stops, the public freeze could be repeated.	Accepted. Only possible when the keeper is down; harmless; monitored.
Info	Info	Some admin changes emitted no events.	Fixed. Events added.

4. What the contracts guarantee

- Money can only end up in a coin's own locked pool, as the published fees, or burned through buybacks; no admin can take it.
- A coin's supply can never exceed 1,000,000,000 across all chains.
- Pool liquidity is locked forever; no function can remove it.
- The winning chain is decided by on-chain math from every chain's report; the keeper cannot choose it.
- Fees have hard caps in the contracts; sells can never be paused.

5. Remaining risks

- Dependence on third-party systems: LayerZero (two independent verifiers), Across, Uniswap v4, and the issuers of USDC and USDG.
- Admin keys: moving to a 2-of-3 multisig in the first week after launch.
- Beta limit: money held by the launch curves is capped per chain at launch and raised gradually.
- Meme coins are highly risky; this review is about the contracts, not about any coin.